STUDENT PERFORMANCE DATA ANALYSIS

Python Data Science Final Project

Dataset: Student Performance (Mathematics)

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn and OOP

Student Name: _____

Department / Institute: _____

1. Dataset Overview

This project analyzes student performance data using Python. The dataset contains student characteristics and academic results.

Number of rows: 395

Number of columns: 33

The target variable is G3, representing the final mathematics grade.

Original dataset source: UCI Machine Learning Repository, Student Performance Data Set.

Source URL: https://archive.ics.uci.edu/dataset/320/student+performance

Data Quality

Missing values in cleaned dataset: 0

Duplicate rows in cleaned dataset: 0

The dataset was explored using head, tail, info and describe.

2. Methodology and Statistical Analysis

Pandas was used to load, inspect, group and summarize the dataset. NumPy was used to calculate numerical statistics and filter grades.

Mean final grade (G3): 10.42

Median final grade (G3): 11.00

Standard deviation: 4.58

Minimum final grade: 0

Maximum final grade: 20

Two classes were implemented: StudentAnalyzer for data summaries and StudentVisualizer for plotting.

3. Research Questions and Findings

Q1. What is the average final grade? The mean is 10.42 and the median is 11.00.

Q2. Do grades differ by gender? Female mean: 9.97; male mean: 10.91. This is a descriptive comparison.

Q3. How does study time relate to grades? The average G3 by study-time category is: {1: 10.05, 2: 10.17, 3: 11.4, 4: 11.26}.

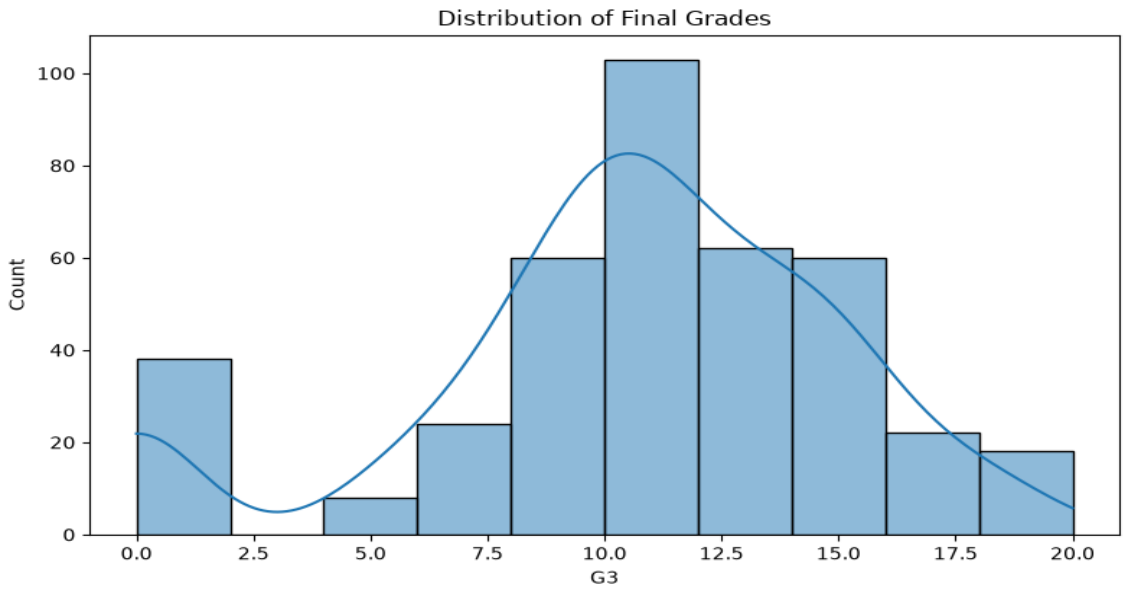
Q4. How are absences related to grades? The correlation between absences and G3 is 0.034.

Q5. Which numerical variable has the strongest absolute correlation with G3 among the other numeric columns? G2 (correlation 0.905).

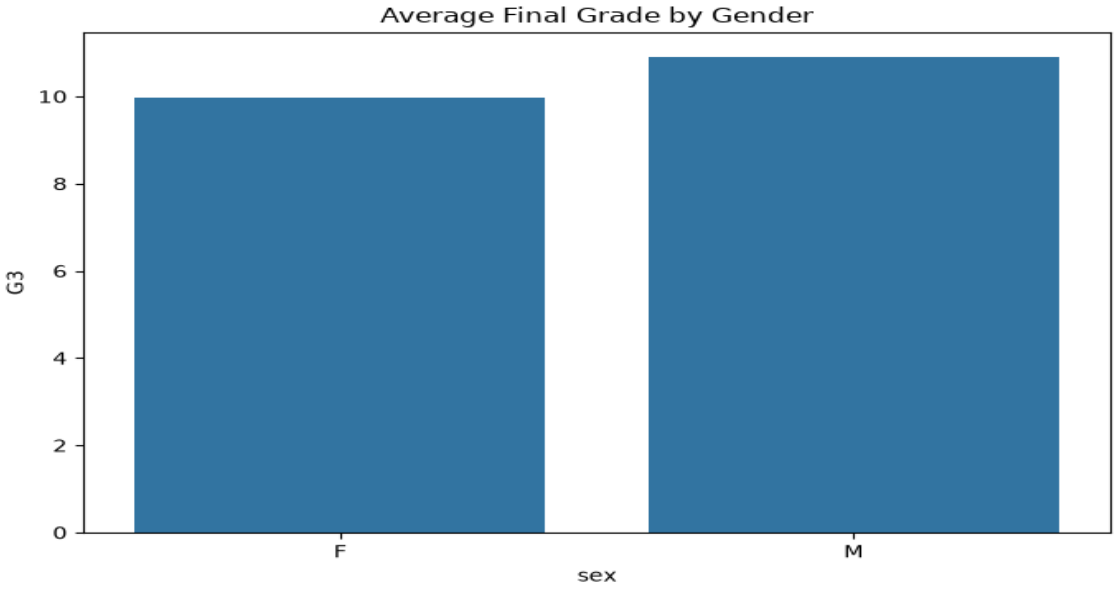
These results describe associations and do not establish causation.

4. Visualizations

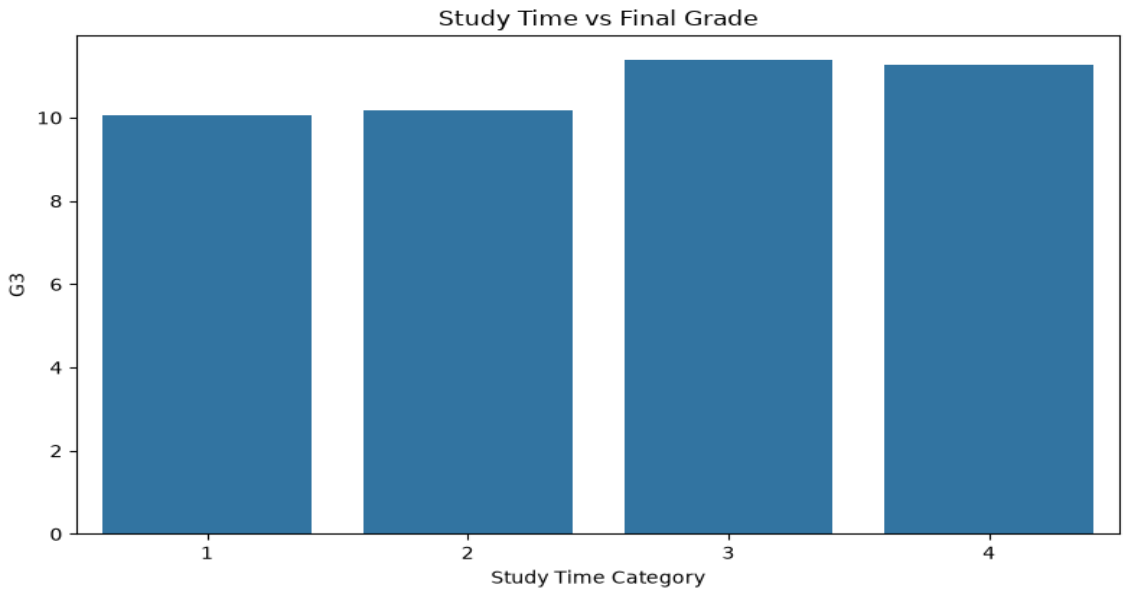
01 grade distribution.png



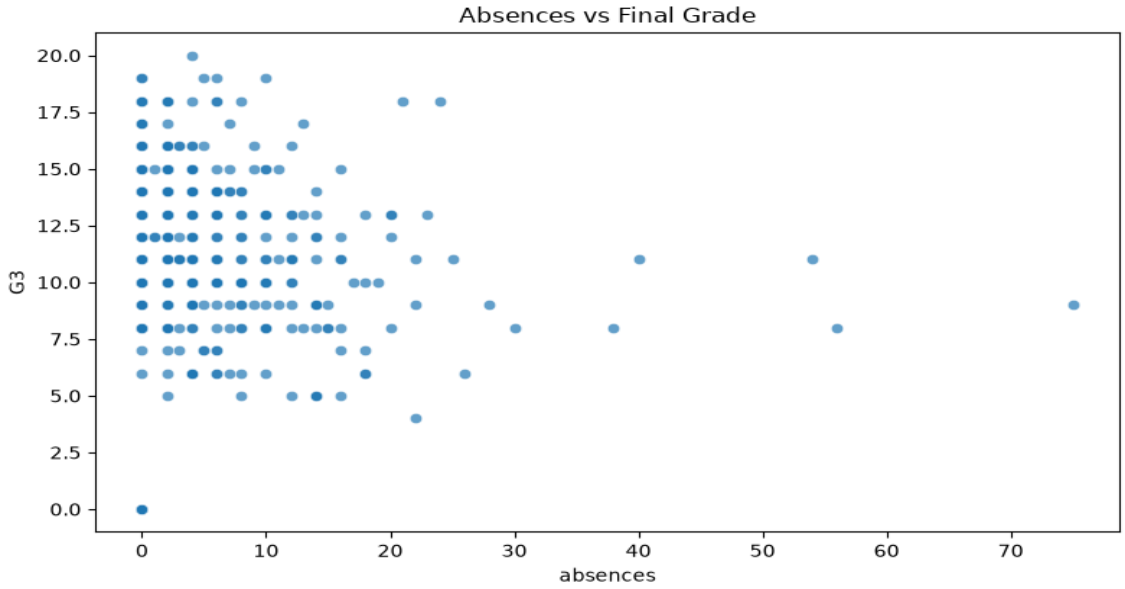
02 grade by gender.png



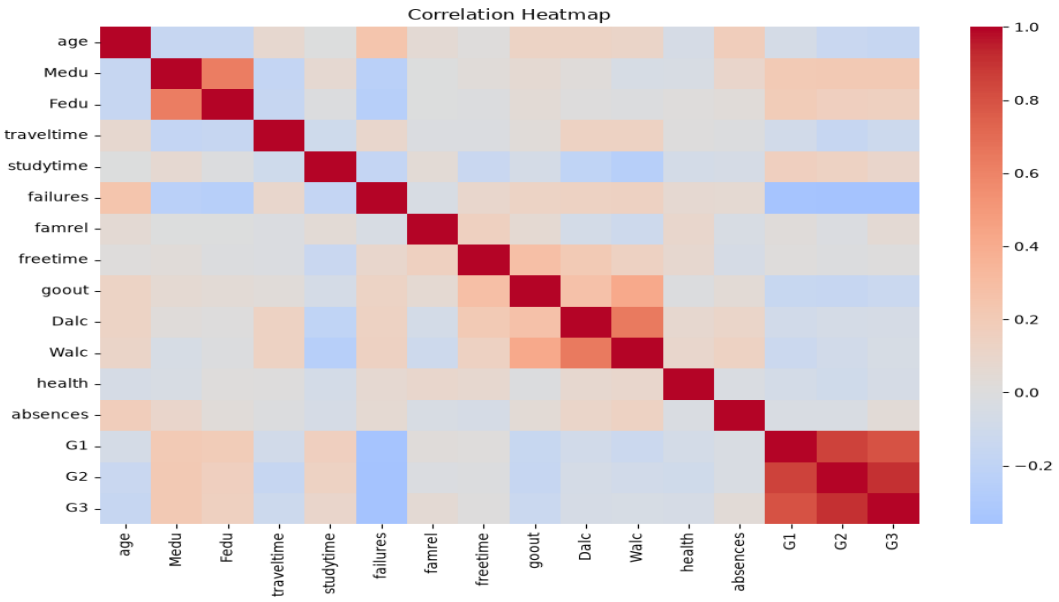
03 studytime vs grade.png



04 absences vs grade.png



05 correlation heatmap.png


Correlation Heatmap
age
Medu
Fedu
traveltime
studytime
failures
famrel
freetime
goout
Dalc
Walc
health
absences
G1
G2
G3
1.0
0.8
0.6
0.4
0.2
0.0
−0.2

5. Limitations

The analysis is based on one dataset and may not represent all students or schools. Correlation does not prove causation. Some important factors may not be included in the dataset.

6. Conclusion

This project demonstrated data loading, exploration, cleaning, numerical analysis, grouping, object-oriented programming and visualization. The findings can help describe patterns in student performance, but should not be treated as proof of cause and effect.

References

Cortez, P. and Silva, A. (2008). Using Data Mining to Predict Secondary School Student Performance.

UCI Machine Learning Repository: https://archive.ics.uci.edu/dataset/320/student+performance